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                                    1        10        20        30        40        50        60
Escherichia_coli_groL               ATGGCAGCTAAAGACGTAAAATTCGGTAACGACGCTCGTGTGAAAATGCTGCGCGGCGTA
Thermus_thermophilus_groL           ATG...GCGAAGATCCTGGTGTTTGACGAGGCTGCGCGCCGCGCCCTGGAGCGCGGGGTC
Bifidobacterium_animalis_groL       ATG...GCAAAGATCATTGAATACGATGAGGAAGCTCGTCAGGGTATGCTCGCCGGTCTG
Corynebacterium_glutamicum_groL     ATG...GCAAAGCTCATTGCTTTTGACCAGGACGCCCGCGAAGGCATTCTCCGGGGCGTT

                                             70        80        90       100       110       120
Escherichia_coli_groL               AACGTACTGGCAGATGCAGTGAAAGTTACCCTCGGTCCAAAAGGCCGTAACGTAGTTCTG
Thermus_thermophilus_groL           AACGCGGTGGCCAACGCGGTGAAGGTCACCCTCGGTCCTCGGGGCCGGAACGTGGTCCTG
Bifidobacterium_animalis_groL       GATAAGCTGGCCGATACCGTCAAGGTGACGCTTGGTCCCAAGGGCCGCAACGTCGTGCTT
Corynebacterium_glutamicum_groL     GACGCTCTGGCAAACGCTGTCAAGGTAACCCTCGGCCCACGCGGCCGTAACGTGGTTCTT

                                            130       140       150       160       170       180
Escherichia_coli_groL               GATAAATCTTTCGGTGCACCGACCATCACCAAAGATGGTGTTTCCGTTGCTCGTGAAATC
Thermus_thermophilus_groL           GAGAAGAAGTTTGGCTCCCCCACCATCACCAAGGACGGGGTGACGGTGGCCAAGGAGGTG
Bifidobacterium_animalis_groL       GACAAGACCTATGGCGCGCCGACCATCACCAACGATGGCGTTTCGATCGCCAAGGAGATC
Corynebacterium_glutamicum_groL     GATAAGGCATTCGGCGGACCTCTGGTCACCAACGACGGTGTCACCATTGCCCGCGACATC

                                            190       200       210       220       230       240
Escherichia_coli_groL               GAACTGGAAGACAAGTTCGAAAATATGGGTGCGCAGATGGTGAAAGAAGTTGCCTCTAAA
Thermus_thermophilus_groL           GAGCTGGAGGACCACCTGGAGAACATCGGGGCCCAGCTCCTCAAGGAGGTGGCCTCCAAG
Bifidobacterium_animalis_groL       GATCTTGAGGATCCGTTCGAGCGCATTGGTGCCGAACTGGTGAAGGAAGTGGCCAAGCGC
Corynebacterium_glutamicum_groL     GACCTTGAGGATCCTTTTGAGAACCTCGGTGCGCAGCTGGTGAAGTCCGTTGCTGTTAAG

                                            250       260       270       280       290       300
Escherichia_coli_groL               GCAAACGACGCTGCAGGCGACGGTACCACCACTGCAACCGTACTGGCTCAGGCTATCATC
Thermus_thermophilus_groL           ACCAACGACGTGGCCGGTGACGGGACCACCACGGCCACCGTCCTCGCCCAGGCCATCGTC
Bifidobacterium_animalis_groL       ACCGATGATGTCGCCGGCGATGGCACCACCACCGCAACTGTACTTGCACAGTCCCTCGTG
Corynebacterium_glutamicum_groL     ACCAACGACATCGCTGGTGACGGCACCACGACTGCAACTCTGCTTGCTCAGGCACTCATT

                                            310       320       330       340       350       360
Escherichia_coli_groL               ACTGAAGGTCTGAAAGCTGTTGCTGCGGGCATGAACCCGATGGACCTGAAACGTGGTATC
Thermus_thermophilus_groL           CGGGAGGGCCTGAAGAACGTGGCCGCCGGGGCCAACCCCCTCGCCCTCAAGCGGGGCATT
Bifidobacterium_animalis_groL       CATGAGGGTCTGAAGAACGTGGTGGCCGGATCCAATCCGATTGCCCTGCGCCGCGGCATC
Corynebacterium_glutamicum_groL     GCTGAAGGCCTGCGCAACGTTGCTGCTGGCGCAAACCCAATGGAGCTCAACAAGGGTATT

                                            370       380       390       400       410       420
Escherichia_coli_groL               GACAAAGCGGTTACCGCTGCAGTTGAAGAACTGAAAGCGCTGTCCGTACCATGCTCTGAC
Thermus_thermophilus_groL           GAGAAGGCGGTGGAGGCCGCGGTGGAGAAGATCAAGGCCCTCGCCATCCCCGTGGAGGAC
Bifidobacterium_animalis_groL       GAGAAGGCTTCCGACGCTCTCGTCAAGCAGCTTGTCGCTTCCGCGAAGCCGGTGGAGACC
Corynebacterium_glutamicum_groL     TCTGCAGCTGCAGAAAAGACCTTGGAAGAGTTGAAGGCACGCGCAACCGAGGTGTCTGAC

                                            430       440       450       460       470       480
Escherichia_coli_groL               TCTAAAGCGATTGCTCAGGTTGGTACCATCTCCGCTAACTCCGACGAAACCGTAGGTAAA
Thermus_thermophilus_groL           CGGAAGGCCATTGAGGAGGTGGCCACCATCTCCGCCAA...CGATCCCGAGGTCGGCAAG
Bifidobacterium_animalis_groL       AAGGAACAGATTGCTGCAACCGCAACGATTTCCGCAGG...TGACCCCGAGGTGGGCGAG
Corynebacterium_glutamicum_groL     ACCAAGGAAATCGCAAACGTCGCTACCGTTTCATCCCG...CGATGAAGTTGTCGGCGAG

                                            490       500       510       520       530       540
Escherichia_coli_groL               CTGATCGCTGAAGCGATGGACAAAGTCGGTAAAGAAGGCGTTATCACCGTTGAAGACGGT
Thermus_thermophilus_groL           CTGATTGCCGACGCCATGGAGAAGGTGGGGAAGGAGGGCATCATCACCGTTGAGGAGTCC
Bifidobacterium_animalis_groL       AAGATCGCCGAGGCTCTCGACAAGGTCGGACAGGATGGCGTCGTCACCGTCGAAGACAAC
Corynebacterium_glutamicum_groL     ATCGTTGCTGCAGCGATGGAAAAGGTTGGCAAGGACGGTGTCGTCACCGTTGAGGAGTCC

                                            550       560       570       580       590       600
Escherichia_coli_groL               ACCGGTCTGCAGGACGAACTGGACGTGGTTGAAGGTATGCAGTTCGACCGTGGCTACCTG
Thermus_thermophilus_groL           AAGAGCCTCGAGACCGAGCTCAAGTTCGTTGAGGGGTACCAGTTTGACAAGGGGTACATC
Bifidobacterium_animalis_groL       AACCGCTTCGGCCTCGATCTCGATTTCACCGAGGGCATGCGCTTCGACAAGGGTTATATT
Corynebacterium_glutamicum_groL     CAGTCCATCGAGACTGCTCTCGAGGTCACCGAAGGTATTTCTTTCGACAAGGGCTACCTT
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                                          610       620       630       640       650       660
Escherichia_coli_groL               TCTCCTTACTTCATCAACAAGCCGGAAACTGGCGCAGTAGAACTGGAAAGCCCGTTCATC
Thermus_thermophilus_groL           TCCCCCTACTTCGTCACCAACCCCGAGACGATGGAGGCGGTCCTCGAGGACGCCTTCATC
Bifidobacterium_animalis_groL       TCCCCCTACTTCGTCACCAATGCGGAAGACCAGACGGCCGTGCTCGACGACCCGTACATT
Corynebacterium_glutamicum_groL     TCCCCTTATTTCATCAACGACAACGACACTCAGCAGGCTGTCCTGGACAACCCTGCAGTG

                                          670       680       690       700       710       720
Escherichia_coli_groL               CTGCTGGCTGACAAGAAAATCTCCAACATCCGCGAAATGCTGCCGGTTCTGGAAGCTGTT
Thermus_thermophilus_groL           CTCATCGTGGAGAAGAAGGTCTCCAACGTCCGTGAGCTCCTCCCCATCCTGGAGCAGGTG
Bifidobacterium_animalis_groL       CTGCTCACCTCGAGCAAGGTCTCCTCGCAGCAGGATGTGGTCCACATCGCCGAACTGGTG
Corynebacterium_glutamicum_groL     CTGCTTGTTCGCAACAAGATTTCTTCCCTCCCAGACTTCCTCCCATTGCTGGAGAAGGTT

                                          730       740       750       760       770       780
Escherichia_coli_groL               GCCAAAGCAGGCAAACCGCTGCTGATCATCGCTGAAGATGTAGAAGGCGAAGCGCTGGCA
Thermus_thermophilus_groL           GCCCAGACGGGCAAGCCCCTCCTGATCATCGCCGAGGACGTGGAGGGCGAGGCTTTGGCC
Bifidobacterium_animalis_groL       ATGAAGACCGGCAAGCCGTTGCTGATCGTGGCCGAGGATGTCGATGGCGAGGCACTGCCC
Corynebacterium_glutamicum_groL     GTGGAGTCCAACCGTCCTTTGCTGATCATCGCAGAAGACGTCGAGGGCGAGCCTTTGCAG

                                          790       800       810       820       830       840
Escherichia_coli_groL               ACTCTGGTTGTTAACACCATGCGTGGCATCGTGAAAGTCGCTGCGGTTAAAGCACCGGGC
Thermus_thermophilus_groL           ACCCTGGTGGTGAACAAGCTCCGGGGCACCCTGAGCGTGGCCGCGGTGAAGGCTCCCGGC
Bifidobacterium_animalis_groL       ACGCTGATTCTCAACAACATCCGTGGCACCTTCAAGTCCTGCGCCGTCAAGGCTCCGGGC
Corynebacterium_glutamicum_groL     ACCCTGGTTGTGAACTCCATCCGCAAGACCATCAAGGTCGTTGCAGTGAAGTCCCCTTAC

                                          850       860       870       880       890       900
Escherichia_coli_groL               TTCGGCGATCGTCGTAAAGCTATGCTGCAGGATATCGCAACCCTGACTGGCGGTACCGTG
Thermus_thermophilus_groL           TTCGGTGACCGCAGGAAGGAGATGCTCAAGGACATCGCGGCCGTCACCGGCGGCACCGTG
Bifidobacterium_animalis_groL       TTCGGTGACCGCCGCAAGGCCATGCTGCAGGATATGGCGATTCTGACCGGCGGTCAGGTC
Corynebacterium_glutamicum_groL     TTCGGTGACCGACGCAAGGCGTTCATGGATGACCTGGCTATTGTCACCAAGGCAACTGTC

                                          910       920       930       940       950       960
Escherichia_coli_groL               ATCTCTGAAGAGATCGGTATGGAGCTGGAAAAAGCAACCCTGGAAGACCTGGGTCAGGCT
Thermus_thermophilus_groL           ATCTCCGAGGAGCTCGGCTTCAAGCTGGAGAACGCCACCCTCTCCATGCTGGGCCGGGCC
Bifidobacterium_animalis_groL       GTCTCGGAGGATCTGGGCCTCAAGCTCGATTCCATCGACCTGTCCGTGTTCGGCACCGCC
Corynebacterium_glutamicum_groL     GTGGATCCAGAAGTGGGCATCAACCTCAACGAAGCTGGCGAAGAAGTTTTCGGTACCGCA

                                          970       980       990      1000      1010      1020
Escherichia_coli_groL               AAACGTGTTGTGATCAACAAAGACACCACCACTATCATCGATGGCGTGGGTGAAGAAGCT
Thermus_thermophilus_groL           GAGCGGGTGCGCATCACCAAGGACGAGACCACCATCGTGGGCGGCAAGGGCAAGAAGGAG
Bifidobacterium_animalis_groL       AAGAAGGTCATTGTCTCCAAGGATGAGACCACCATCGTGTCCGGTGGCGGCTCCAAGGAG
Corynebacterium_glutamicum_groL     CGCCGCATCACCGTTTCCAAGGACGAAACCATCATCGTTGATGGTGCAGGTTCCGCAGAA

                                         1030      1040      1050      1060      1070      1080
Escherichia_coli_groL               GCAATCCAGGGCCGTGTTGCTCAGATCCGTCAGCAGATTGAAGAAGCAACTTCTGACTAC
Thermus_thermophilus_groL           GACATTGAGGCCCGGATCAACGGCATCAAGAAGGAGCTGGAGACCACCGACAGCGAGTAC
Bifidobacterium_animalis_groL       GACGTGGCCGCACGCGTCGCCCAGATTCGCGCCGAGATCGAGAAGACCGATTCCGATTAC
Corynebacterium_glutamicum_groL     GACGTTGAAGCACGTCGCGGCCAGATCCGTCGCGAAATCGCCAACACCGATTCCACCTGG

                                         1090      1100      1110      1120      1130      1140
Escherichia_coli_groL               GACCGTGAAAAACTGCAGGAACGCGTAGCGAAACTGGCAGGCGGCGTTGCAGTTATCAAA
Thermus_thermophilus_groL           GCCCGGGAGAAGCTCCAGGAGCGCCTGGCGAAGCTCGCGGGTGGCGTGGCCGTGATCCGG
Bifidobacterium_animalis_groL       GATCGTGAGAAGCTGCAGGAGCGTTTGGCTAAGCTGGCAGGTGGCGTCGCCGTCATCAAG
Corynebacterium_glutamicum_groL     GATCGCGAAAAGGCAGAAGAGCGTTTGGCTAAGCTCTCCGGTGGTATTGCTGTCATCCGC

                                         1150      1160      1170      1180      1190      1200
Escherichia_coli_groL               GTGGGTGCTGCTACCGAAGTTGAAATGAAAGAGAAAAAAGCACGCGTTGAAGATGCCCTG
Thermus_thermophilus_groL           GTGGGCGCCGCCACGGAGACCGAGCTCAAGGAGAAGAAGCACCGCTTTGAGGACGCCCTG
Bifidobacterium_animalis_groL       GTCGGCGCAGCCACCGAGGTCGAGGCCAAGGAGCGCAAGCACCGCATTGAAGATGCCGTG
Corynebacterium_glutamicum_groL     GTTGGTGCAGCAACTGAAACCGAAGTCAACGACCGCAAGCTGCGTGTCGAAGATGCCATC
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                                          1210      1220      1230      1240      1250
Escherichia_coli_groL               CACGCGACCCGTGCTGCGGTAGAAGAAGGCGTGGTTGCTGGTGGTGGTGTTGCGCTGAT.
Thermus_thermophilus_groL           AACGCCACCCGGGCCGCGGTGGAGGAGGGCATCGTGCCCGGCGGCGGCGTGACCCTCCT.
Bifidobacterium_animalis_groL       CGCAACGCCAAGGCCGCCATCGAAGAGGGTCTGGTTCCGGGCGGCGGCGTCGCACTGGTG
Corynebacterium_glutamicum_groL     AACGCTGCTCGCGCAGCAGCACAAGAAGGCGTTATCGCTGGTGGCGGTTCCGCTTTGGTT

                                 1260         1270      1280       1290       1300      1310
Escherichia_coli_groL               CCG...CGTAGCGTCTAAACTGGCTGACCT.GCGTGGT.CAGAACGAAGACCAGAACGTG
Thermus_thermophilus_groL           CCGGGCCATCAGCGCCGTGGAGGAGCTCATCA.AGAAGCTG.GAGGGCGACGAGGCCACG
Bifidobacterium_animalis_groL       CAG.GCTGCCGAAAAGGTTGAGAAGGACTTCA.AC.C..TT.GAAGGCGACGAGGCCACC
Corynebacterium_glutamicum_groL     CAG.ATCGCTGAGACTCTGAAGGCTTACGCCGAAG.AG.TTCGAAGGCGACCAGAAGGTC

                                      1320         1330       1340      1350      1360      1370
Escherichia_coli_groL               GGTATCAA...AGTTGCACTG.CGTGCAATGGAAGCTCCGCTGCGTCAGATCGTATTGAA
Thermus_thermophilus_groL           GGCGCCAA...GATCGTGCGC.CGGGCCCTGGAGGAACCCGCCCGCCAGATCGCCGAGAA
Bifidobacterium_animalis_groL       GGTGCCGC...AATCGTCTTCTCG.GGCATCGAGGCTCCGATTAAGCAGATCGCCGAAAA
Corynebacterium_glutamicum_groL     GGCGTTCGCGCACTGG.CTAC..T.GCTTTGGGCAAGCCAGCGTACTGGATCGCCTCCAA

                                          1380         1390      1400       1410      1420
Escherichia_coli_groL               CTGCGGCGAAGAACCGTC...TGTTGTTGCTAACAC.CGTTAAAGGCGGCGACGGCAA..
Thermus_thermophilus_groL           CGCCGGGTACGAGGCTCCCGTCATCGTCCAGCAGATCCTGGCCGAGA....CCAAGAACC
Bifidobacterium_animalis_groL       TGCAGGTCTCTCTGGCGCCGTGGTGATCGACAAGGTTCGTTCCCTGC....CTGAGGG..
Corynebacterium_glutamicum_groL     CGCAGGTCTTGACGGCTCTGTTGTTGTTGCACGCACTGCTGCTCTGC....CAAACGG..

                                          1430      1440      1450      1460      1470      1480
Escherichia_coli_groL               ....CTACGGTTACAACGCAGCAACCGAAGAATACGGCAACATGATCGACATGGGTATCC
Thermus_thermophilus_groL           CCCGCTACGGCTTCAACGCCGCCACCGGGGAGTTCGTGGACATGGTGGAGGCGGGCATCG
Bifidobacterium_animalis_groL       ....TGAGGGCTTCAATGCGGCAACCGACACCTATGAGGATCTCATGGCCGCCGGTGTGA
Corynebacterium_glutamicum_groL     ....CGAGGGCTTCAACGCTGCAACTTTGGAATACGGAAACCTGATCAACGACGGTGTCA

                                          1490      1500      1510      1520      1530      1540
Escherichia_coli_groL               TGGATCCAACCAAAGTAACTCGTTCTGCTCTGCAGTACGCAGCTTCTGTGGCTGGCCTGA
Thermus_thermophilus_groL           TTGACCCCGCCAAGGTGACCCGCTCTGCCCTGCAGAACGCCGCCTCCATCGGCGCCCTCA
Bifidobacterium_animalis_groL       CCGATCCTGTCAAGGTGACTCGTTCTGCTTTGCAGAATGCGGCCTCCATCGCGGGTCTGT
Corynebacterium_glutamicum_groL     TCGACCCAGTCAAGGTCACCCATTCCGCAGTAGTGAATGCAACCTCTGTTGCACGCATGG

                                          1550      1560      1570      1580      1590
Escherichia_coli_groL               TGATCACCACCGAATGCATGGTTACCGACCTGCCGAAAAACGATGCAGCTGA.CTTAGGC
Thermus_thermophilus_groL           TCCTCACCACCGAGGCGGTGGTGGCGGAGAAGCCCGAGAAGA.AGGAGTCCACCCC...C
Bifidobacterium_animalis_groL       TCCTGACGACCGAAGCGGTTGTCGCCAACAAGCCGGAGCCGGCAGCGGCA.CCGGCCG..
Corynebacterium_glutamicum_groL     TTCTGACCACTGAGGCTTCTGTTGTTGAGAAGCCTGCAGAAGAAGCAGCCGA.TGCACAT

                                 1600      1610      1620      1630      1640
Escherichia_coli_groL               GCTGCTGGCGGTATGGGCGGCATGGGTGGCATGGGCGGCATGATGTAA
Thermus_thermophilus_groL           G...CCTCCG...CGGG...CGCCGGGGACATGGACTTC......TAA
Bifidobacterium_animalis_groL       ........C...........CGGCCAGGACATGGGTTAC......TGA
Corynebacterium_glutamicum_groL     GC..AGGA............CATCATC.........ACCA....CTAA
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